
Dataset Cartography: Mapping and Diagnosing Datasets with Training Dynamics

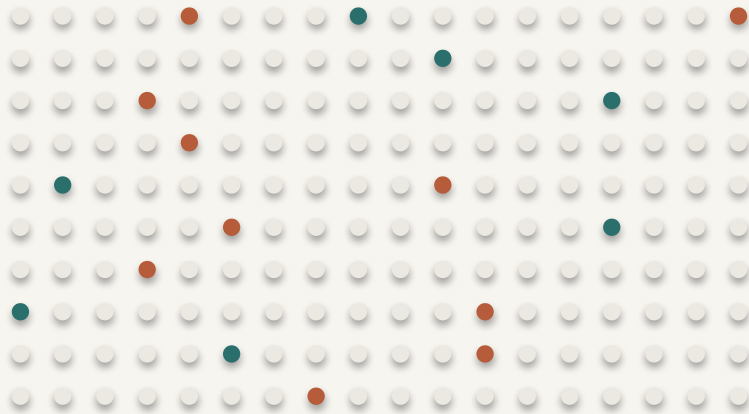
Reproduction of Swayamdipta et al., EMNLP 2020

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RoBERTa-large · SNLI · NLI Diagnostics

Aggregate metrics hide example-level structure

10,000 training examples



Accuracy

92.0%

Loss

0.21

These summaries say little about which examples were learned immediately, learned unreliably, or never learned.

Can we diagnose a dataset by watching how the model learns each example?

SNLI is a three-way natural-language inference task

PREMISE

“Cafe Express” sign covered in graffiti.

ENTAILMENT

A sign covered in graffiti.

must be true

NEUTRAL

The sign is hard to read.

may be true

CONTRADICTION

The sign is brand new and clean.

cannot be true

The model receives a premise–hypothesis pair and predicts their logical relation.

Training dynamics record the path, not only the endpoint

Ordinary analysis

What did the final
model predict?

Epoch 6

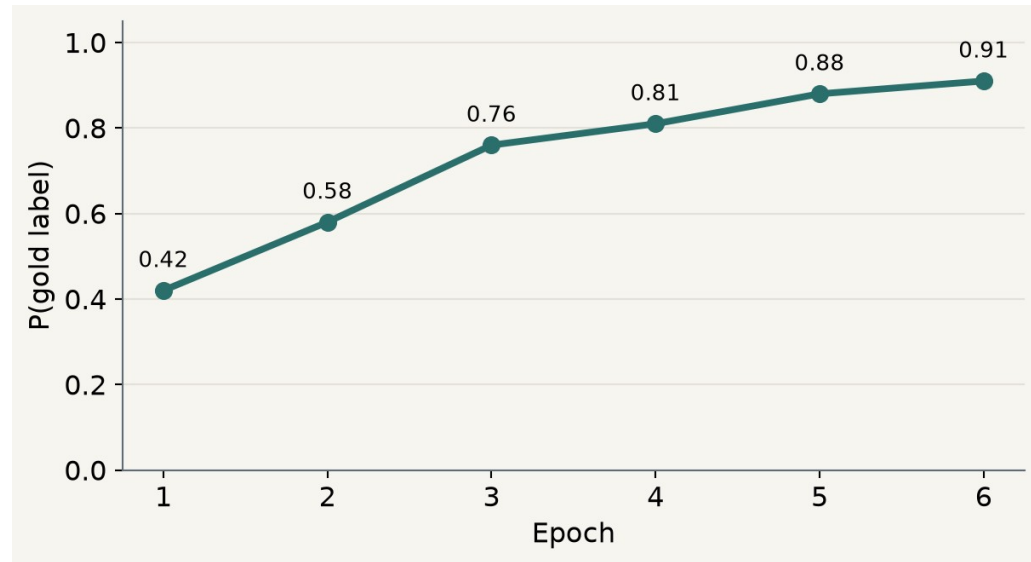
GOLD LABEL

0.91

one decision

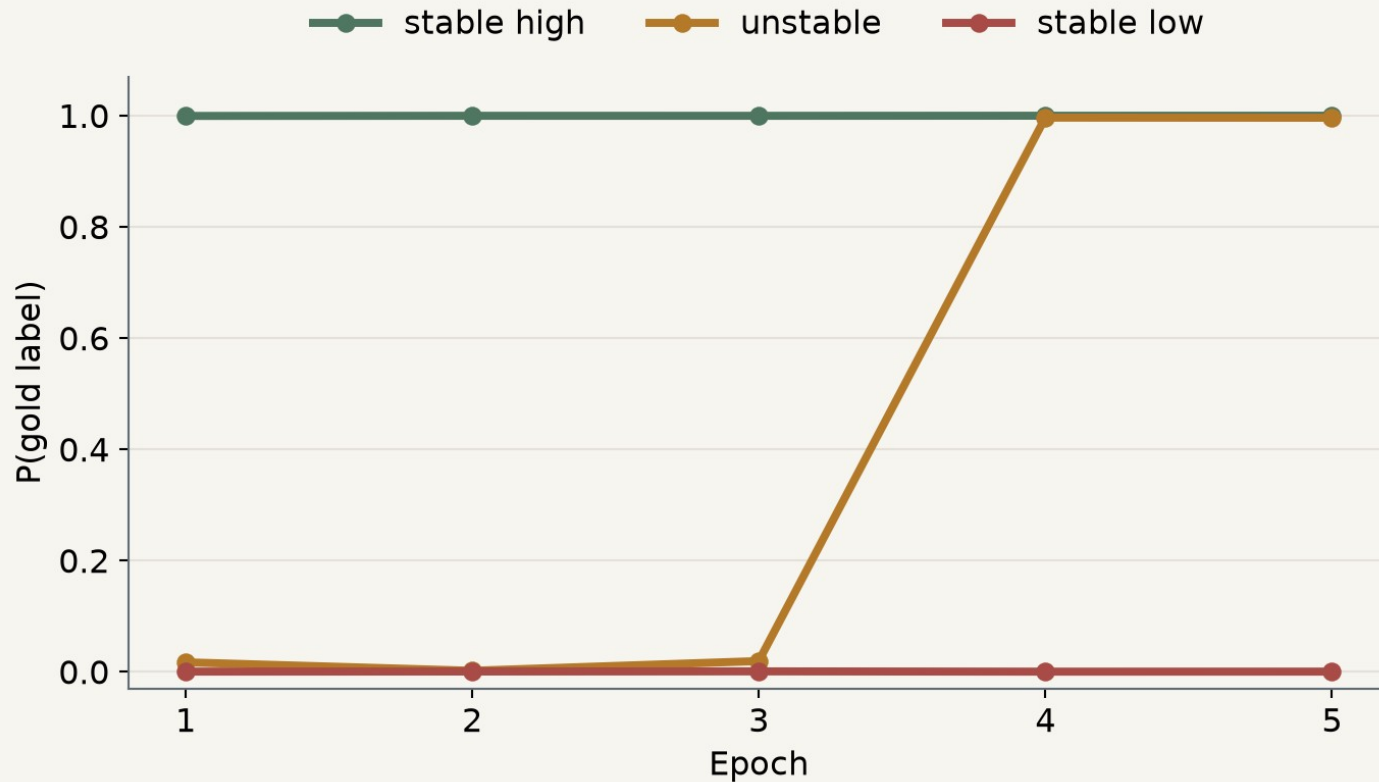


Dataset Cartography



a trajectory across training

Three examples produce three distinct trajectories



Stable high

Beach children → men in a jungle

CONTRADICTION

Unstable

Woman playing tennis → woman practices tennis

CONTRADICTION

Stable low

Girl photographed → boy photographed

ENTAILMENT

How can these trajectories be summarized with a few comparable numbers?

Confidence summarizes how strongly an example is learned

$$\mu_i = \frac{1}{E} \sum_{e=1}^E p_{\theta_e}(y_i | x_i)$$

Average probability assigned to the gold label throughout training.

WHERE

E number of epochs

y_i gold label

p_{θ_e} model probability at epoch e

HIGH CONFIDENCE

The model generally supports the provided label.

LOW CONFIDENCE

The model repeatedly struggles with that label.

Confidence is relative to the gold label and to this model and training procedure.

Variability captures instability; correctness counts decisions

$$\sigma_i = \sqrt{\frac{1}{E} \sum_{e=1}^E (p_{\theta_e}(y_i | x_i) - \mu_i)^2}$$

Standard deviation of gold-label confidence across epochs.

CORRECTNESS

Fraction of epochs in which the predicted class equals the gold label.

stable high confidence



low variability

unstable confidence



high variability

stable low confidence



low variability

Variability can be low when the model is consistently right or consistently wrong.

Correctness helps distinguish those cases.

A data map places every example by confidence and variability



EASY-TO-LEARN

high confidence
low variability

AMBIGUOUS

high variability
confidence may vary

HARD-TO-LEARN

low confidence
often low correctness

Every point is one training example.

Map regions become interpretable through real examples

EASY-TO-LEARN

PREMISE

Two children are on the beach and play in wet sand.

HYPOTHESIS

Two men are walking through a jungle.

CONTRADICTION

confidence 1.000 · variability < .001

AMBIGUOUS

PREMISE

A woman in red and white clothing is playing tennis.

HYPOTHESIS

Dressed all in white, the woman practices tennis.

CONTRADICTION

confidence .406 · variability .482

HARD-TO-LEARN

PREMISE

Photographers photograph a girl sitting in a street.

HYPOTHESIS

The photographer is taking a picture of a boy.

ENTAILMENT

confidence < .001 · variability < .001

Ambiguous ≠ mislabeled. Hard-to-learn ≠ automatically wrong.

The paper tested whether map regions predict useful behavior

SNLI

natural-language inference

MultiNLI

multi-genre inference

WinoGrande

commonsense reasoning

QNLI

question answering
reformulated as NLI

DATA USEFULNESS

Which regions support ID and OOD generalization?

OPTIMIZATION

Can a model learn from only ambiguous examples?

ANNOTATION QUALITY

Are low-confidence regions enriched for problems?

UNCERTAINTY

How do dynamics relate to human agreement?

The maps were hypotheses about dataset structure; these experiments tested their meaning.

Ambiguous subsets often supported stronger generalization

WinoGrande — 33% subsets

Training data	ID	OOD
Full 100%	79.7	86.0
Random	73.3	85.6
Hard-to-learn	77.9	87.2
Ambiguous	78.7	87.6

SNLI — 33% ambiguous

IN-DISTRIBUTION

92.2%

within 0.2 pp of full data

NLI DIAGNOSTICS

63.5%

Across these experiments, challenging and variable examples could preserve ID accuracy while improving independently collected OOD performance.

A recurring empirical pattern — not a universal law about every dataset or model.

Easy examples can be necessary for optimization

WinoGrande: only the most ambiguous data

25–50%

learns

$\leq 17\%$

chance-level

With 17% ambiguous data, replacing only one tenth with easy examples restored learning and beat the random baseline on ID accuracy.

Different regions can serve different roles

AMBIGUOUS

informative decision boundaries

EASY

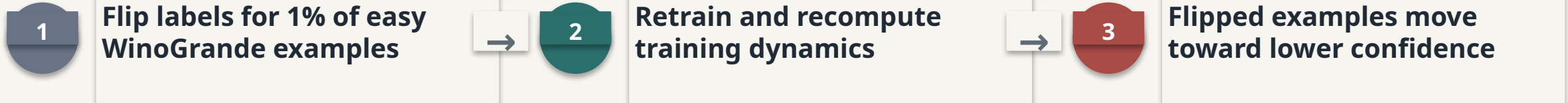
stable optimization signal

TOO MUCH EASY

weaker OOD performance

“Ambiguous good, easy bad” would misstate the paper.

Hard-to-learn regions are enriched for annotation problems



Human reannotation of predicted groups

Dataset	Predicted noisy	Predicted clean
WinoGrande	67%	13%
SNLI	76%	4%

Responsible interpretation

- Low confidence is a useful review signal.
- The region also contains legitimate difficult cases.
- Hard-to-learn does not mean mislabeled.

Data maps prioritize inspection; they do not replace annotation review.

The selected SNLI experiment tests data efficiency and generalization

IN-DISTRIBUTION (ID)

SNLI test set

Same collection process and task distribution as training.

OUT-OF-DISTRIBUTION (OOD)

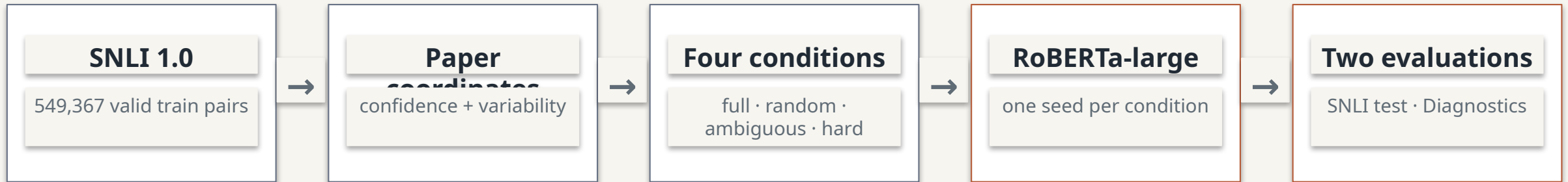
NLI Diagnostics

1,104 hand-crafted examples probing lexical semantics, logic, predicate-argument structure, and knowledge.

Training data	Examples used	SNLI test	Diagnostics
Full	100%	92.0	61.8
Random	33%	91.3	60.4
Hard-to-learn	33%	91.8	62.0
Ambiguous	33%	92.2	63.5

Question: can one third of SNLI retain ID performance and improve diagnostic generalization?

Reproduction protocol follows the published SNLI selection



Selection definitions

- Random: published 33% sampling procedure
- Ambiguous: highest variability
- Hard-to-learn: lowest confidence

Training settings

- learning rate 1.07086×10^{-5}
- effective batch 96 · maximum length 128
- up to 6 epochs · best validation checkpoint

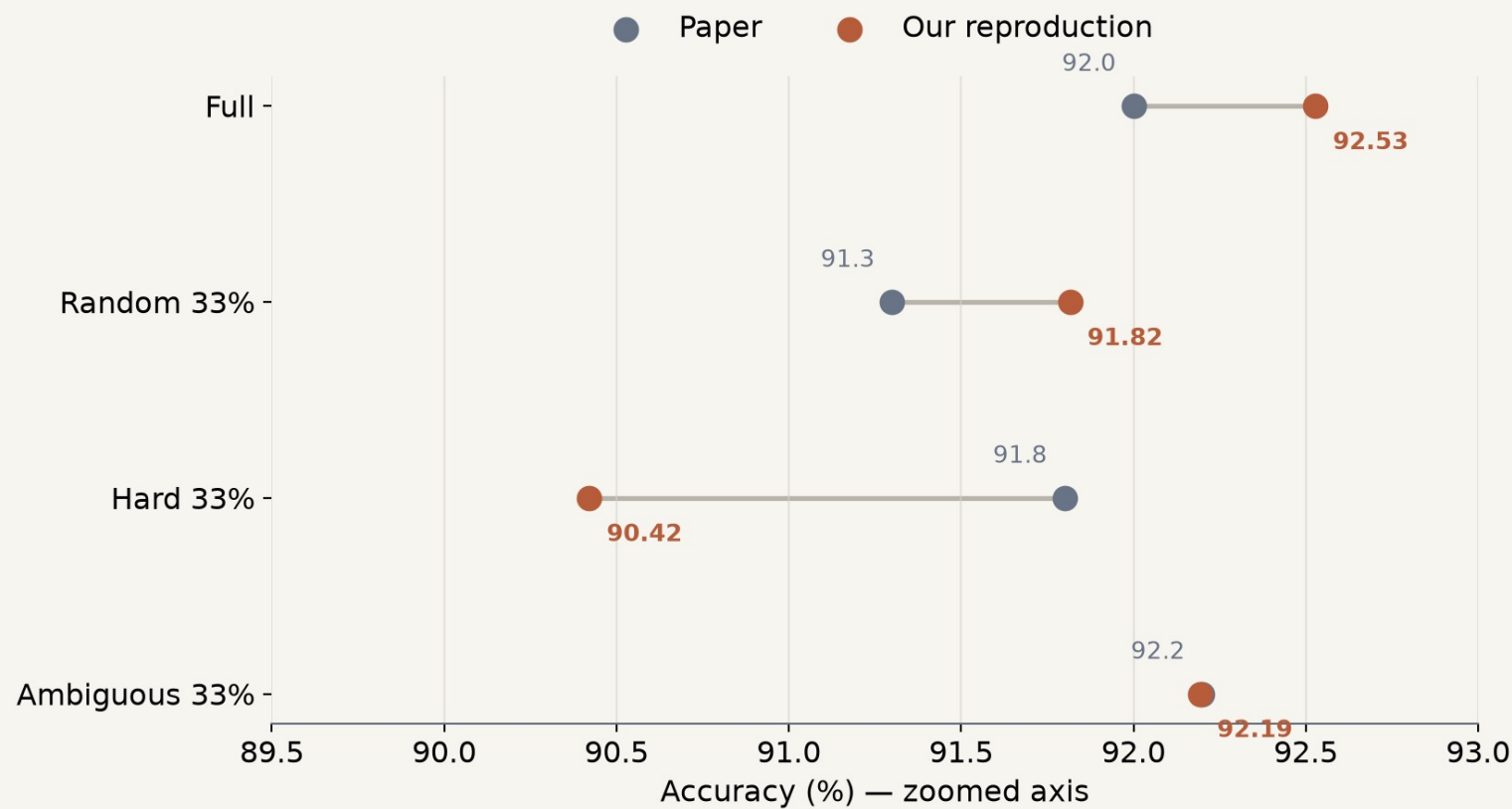
Published coordinates isolate the data-selection claim from map reconstruction differences.

Hardware constraints changed implementation details

Aspect	Original paper	Our reproduction
GPU	Quadro RTX 8000	RTX 4070 Ti SUPER · 16 GB
Batch	physical batch 96	32 × 3 gradient accumulation = 96
Precision	2020 training stack	BF16 · modern PyTorch/Transformers
Padding	length 128	dynamic padding · max length 128
Reporting	best of 3 seeds	one available seed: 93078
Epochs	paper says 6; config early stopping	best checkpoints; 4–6 epochs

These facts make exact equality unlikely while leaving the subset comparison scientifically interpretable.

In-distribution accuracy closely matches the paper



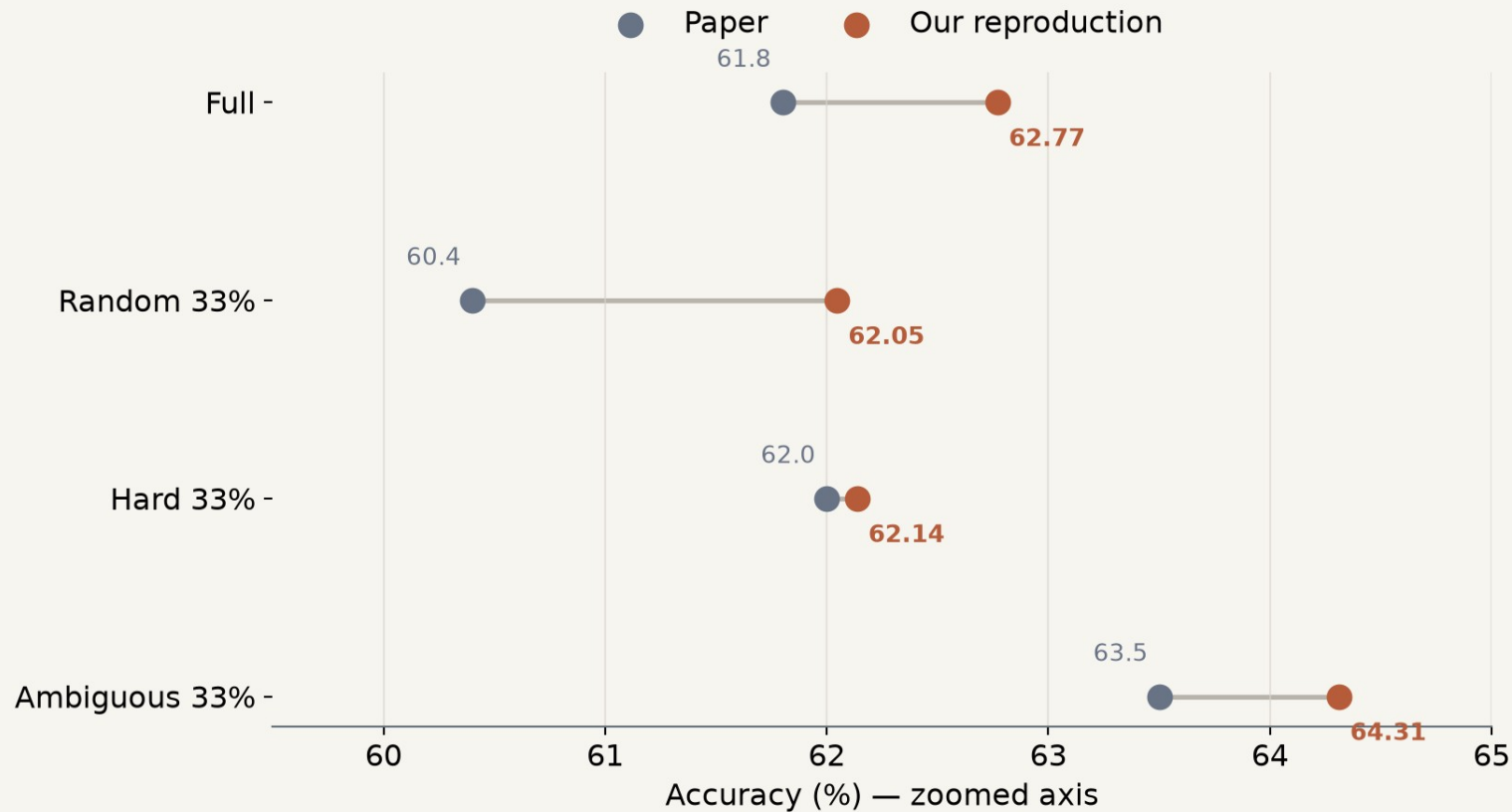
Subset	Paper	Ours	Δ
Full	92.0	92.53	+0.53
Random	91.3	91.82	+0.52
Hard	91.8	90.42	-1.38
Ambiguous	92.2	92.19	-0.01

Central reproduction result

Ambiguous: 92.19 vs 92.2

Axis is deliberately labeled as zoomed; direct values and differences are shown.

The ambiguous subset also leads on NLI Diagnostics



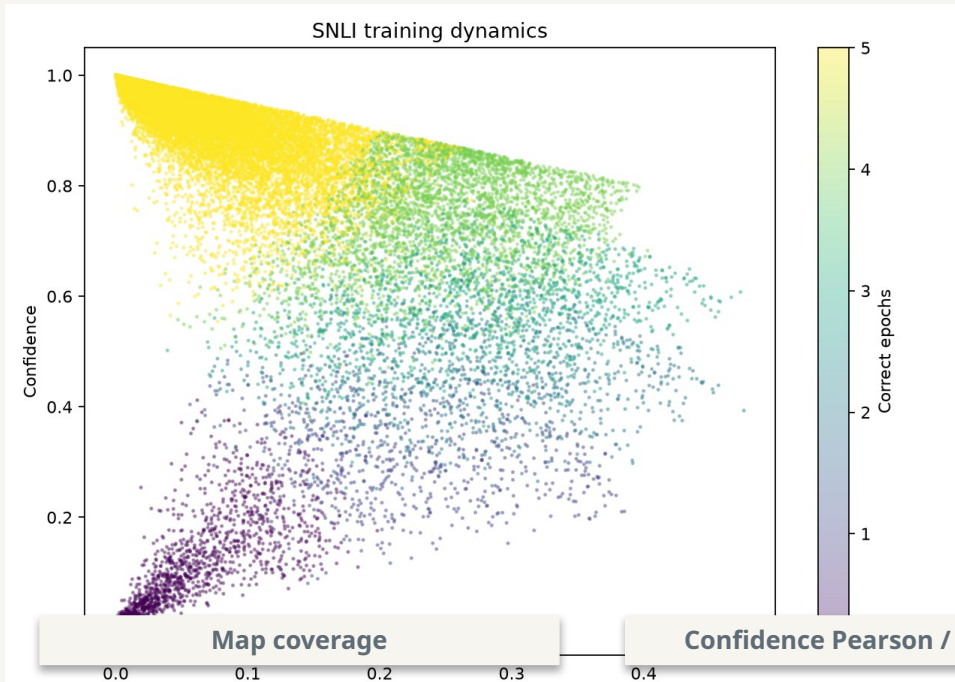
Subset	Paper	Ours	Δ
Full	61.8	62.77	+0.97
Random	60.4	62.05	+1.65
Hard	62.0	62.14	+0.14
Ambiguous	63.5	64.31	+0.81

Our Diagnostics ranking

Ambiguous > Full
> Hard > Random

NLI Diagnostics contains 1,104 labeled examples; one seed can move several examples.

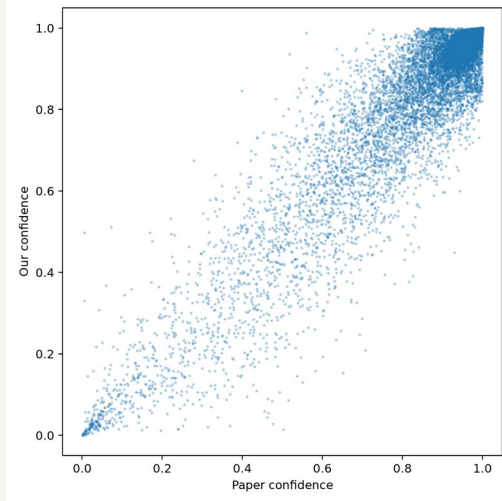
Reproduced maps agree more on confidence than variability



99.91%

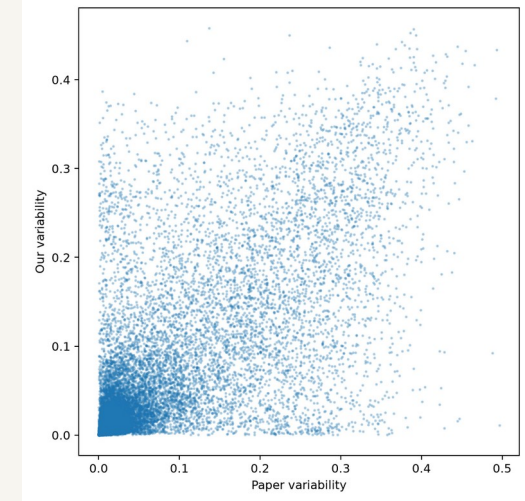
Confidence Pearson / Spearman

0.944 / 0.888



Variability Pearson / Spearman

0.708 / 0.772



Ambiguous-set overlap

75.2%

Individual agreement: the paper's "girl → boy" hard example is also among our lowest-confidence examples.

Coverage excludes 498 examples lacking exactly one observation in every logged epoch; no values were imputed.

The selection result reproduces; map details remain sensitive

SUPPORTED BY OUR EVIDENCE

- Ambiguous 33% matches paper ID accuracy: 92.19 vs 92.2.
- Ambiguous remains best on Diagnostics: 64.31.
- Confidence structure transfers strongly between maps.
- One third of the data approaches full-data performance.

LESS STABLE OR DIFFERENT

- Hard-subset ID is 1.38 points below the paper.
- Variability agrees less strongly than confidence.
- The paper reports best of 3 seeds; we have one.
- Hardware, precision, and software stacks differ.

Training histories expose structure that final accuracy alone cannot show.

In this SNLI reproduction, selecting high-variability examples preserved accuracy and strengthened diagnostic generalization, while exact map coordinates remained sensitive to training details.

Exact reproduction hyperparameters

Setting	Value
Model	roberta-large
Seed	93078
Learning rate	$1.0708609960508476 \times 10^{-5}$
Optimizer	AdamW; $\epsilon=1\text{e-}8$; weight decay 0; max grad norm 1
Schedule	linear decay; zero warmup
Maximum length	128 tokens
Batch	32 physical $\times$ 3 accumulation = 96 effective
Precision	BF16
Checkpoint	highest SNLI validation accuracy
Epoch cap	6; patience 3

Individual run metrics

Run	Train N	Epochs	Runtime	Validation	SNLI test	Diagnostics
Full	549,367	5.00	120.0 min	93.23	92.53	62.77
Random 33%	181,291	5.00	40.6 min	92.40	91.82	62.05
Hard 33%	182,335	4.00	33.9 min	91.21	90.42	62.14
Ambiguous 33%	182,335	6.00	50.5 min	93.05	92.19	64.31

All runs use one seed. Epoch counts differ because early stopping selected the best validation checkpoint.

Raw JSON files: results/run_metrics/

Subset construction

RANDOM 33%

181,291 examples

Pandas sample with the recovered subset seed 725862.

AMBIGUOUS 33%

182,335 examples

Sort released coordinates by highest variability.

HARD-TO-LEARN 33%

182,335 examples

Sort released coordinates by lowest confidence.

FULL

549,367 examples

Every official SNLI training example with a valid label.

All coordinate joins are one-to-one and cover every valid training ID before selection.

Complete paper-versus-reproduction table

Subset	Paper ID	Our ID	Δ ID	Paper OOD	Our OOD	Δ OOD
Full	92.0	92.53	+0.53	61.8	62.77	+0.97
Random 33%	91.3	91.82	+0.52	60.4	62.05	+1.65
Hard 33%	91.8	90.42	-1.38	62.0	62.14	+0.14
Ambiguous 33%	92.2	92.19	-0.01	63.5	64.31	+0.81

Paper values are best-of-three; reproduction values are one run with seed 93078.

Reproduction environment and artifact locations

Environment

- NVIDIA RTX 4070 Ti SUPER · 16 GB
- Python 3.12 · PyTorch 2.3
- Transformers 4.44–4.49 range
- CUDA · BF16 · dynamic padding

Repository artifacts

- results/reproduction_metrics.csv
- results/run_metrics/*.json
- results/map_comparison.json
- presentation/generate_presentation.py
- presentation/speaker_notes.md

The deck can be regenerated without model checkpoints from the committed compact results and assets.